

5CS037 – Concepts & Technologies of AI

Large Language Models (LLMs) and Education

Name: UGESH DHUNGEL

Group: L5CG3

Student ID: 2513636

Module Leader : Ayush regmi

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Abstract

LLMs are increasingly reshaping educational environments through the delivery of personalized learning experiences, automated tutoring, and intelligent assessment systems. Although such new technologies provide a unique opportunity to increase accessibility, efficiency, and learning support, they equally involve serious ethical issues that cannot be underestimated. Concerns linked to bias, accuracy, transparency, data privacy, and the over automation of teaching roles face the responsible use of LLMs in education. This report develops a critical analysis of ethical implications involving LLM generated educational content, AI-based tutoring, and automated assessment systems. It further discusses relevant international ethical AI initiatives and develops a generic ethical AI framework to ensure fairness, accountability, and human oversight in educational contexts.

Introduction

The spate of recent developments in the sector of Artificial Intelligence (AI), more specifically, the variant of AI known as Large Language Models (LLMs), has created a phenomenal impact on the world of education. The potential of these LLMs, consisting of models like deepseek and other variants, to perform like human beings and supply text responses to queries and answers, and their potential to deliver feedback, has made it extremely attractive to be used on the education sector.

However, the use of LLMs in the sector has raised pertinent and intricate questions about ethics and morality in implementation these technologies in the educational sector. Firstly, there is an issue of the validity of educational content offered by AI. This is because LLMs would produce incorrect knowledge. Second, there is equity in situations where AI founded establishments have incorrect databases that would inhibit other learners. Thirdly, there is an issue of privacy, especially when these establishments get hold of private data of learners.

Fairness, Transparency, Accountability, Human Review, and Privacy as AI ethics can be ensured in order to guarantee the use and impact of LLMs on the education field remain beneficial and do not have any negative effects on it. Also, there exist some regulatory norms such as the UNESCO AI Ethics Recommendation, OECD AI Principles, and EU AI Act, which can be used as a guide for ensuring fairness in AI.

Bias and Accuracy in Educational Content

One of the biggest concerns in the use of LLMs in the educational sector is biases and accuracy in the use of LLMs in producing educational materials. LLMs are trained using large amounts of data drawn from books, articles, and the Web, which may have various biases, including those based on Cultures, Gender, and Social economic. This results in biased results from the use of LLMs in generating an explanation.

Inaccurate data this is another problem. This is because LLMs base their answers on probabilities and may come up with inaccurate explanations, made up quotes and oversimplified responses. This can be problematic in the classroom because it can have a negative impact on the performance of the pupils. Even more problematic is the fact that the pupils may not be able to check it.

To counter these challenges, educational institutions need to establish human review processes, encourage diversity and Representative training data, and foster Critical AI literacy transparency regarding the limitations of AI is the key to maintaining trust and quality in education.

Automation of Tutoring and Feedback

LLMs are extensively applied in the field of artificial tutors in AI that give immediate feedback and guiding instructions. Such platforms could increase convenience, especially for students who do not receive individualized academic attention. But the overreliance on automated tutors raises moral issues of diminishing the involvement of human teachers.

First of all, educators are more than just knowledge deliverers; they are also people's mentors, motivators, and emotional inspirers too. The use of AI tutors may affect students negatively with respect to their socialization, critical thinking, and motivation for more knowledge acquisition. Also, AI does not know the needs of a person emotionally or contextually.

It is important to keep in mind the ethical use of LLMs as far as human control is concerned. AI must always function as an aid to teachers rather than as a replacement. AI can complement traditional teaching to achieve hybrid models where AI is relied upon for mundane tasks.

AI-Based Assessment and Grading

AI-based evaluation tools, fueled by the power of LLMs, are already employed at different levels such as assignment grading, feedback, and evaluating performance. Although such technologies increase the efficiency of the procedure, there remain concerns related to their accountability.

Computerized grading can disadvantage some students whose writing patterns, language skills, and cultural expressions may not match the data sets used in training the algorithm. This is aside from the point that AI grading solutions can form "black boxes," through which students cannot easily justify grades even if they have been incorrectly awarded.

AI-based testing should incorporate responsibilities such as explainability, appeal, and bias audits. Human oversight of the grading process is crucial for accountability and upholding the academic rights of students.

Ethical AI Initiatives and Proposed Framework

International initiatives such as UNESCO's Recommendation Concerning the Ethics of Artificial Intelligence, OECD Principles on Artificial Intelligence, and the EU AI Act recommend responsible use of AI in education. The key recommendations in these initiatives include fairness, transparency, human control, and data protection.

Proposed AI Ethics Framework for LLMs in Educations:

Fairness: Biases Monitoring and Comprehensive Training Data

Transparency: Transparency of Explainable AI results and visibility into the use of ai

Human Oversight: The teacher maintains control

Accountability: Accountability is the state where one is responsible for the actions

Privacy: Safe Data Management and Informed Consent.

Discussion / Personal Reflection

This research highlights the importance of ethical considerations when integrating LLMs into education. While LLMs offer powerful opportunities to enhance learning accessibility, personalization, and efficiency, they also introduce risks that cannot be ignored. I learned that bias and inaccuracy in AI-generated content can significantly impact students, particularly those from marginalized backgrounds. The study also reinforced the idea that education is not purely a technical process but a human-centered activity that requires empathy, mentorship, and ethical judgment.

Ethical AI matters because educational systems shape future generations. Unethical use of AI could deepen inequality, reduce trust in education, and harm student development. Responsible AI frameworks that emphasize fairness, transparency,

and human oversight can guide safer and more inclusive AI deployment. In the long term, ethical AI has the potential to support equitable education, while unethical practices may undermine academic integrity and social trust. Therefore, balancing innovation with ethical responsibility is essential for sustainable educational progress.

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